

47 Arc length

Definition 47.1.

Let C be a C^1 curve in $\mathbb{R}^k$ given parametrically by $\mathbf{g}$,
that is, $\mathbf{g} : [a, b] \rightarrow \mathbb{R}^k$ is C^1 and injective on $[a, b]$ such that $C = \{\mathbf{g}(t) \in \mathbb{R}^k : a \leq t \leq b\}$.
Let $\mathbf{r}$ and $\mathbf{r} + d\mathbf{r}$ be two nearby points in C .

We denote $d\mathbf{r} := (dx_1, \dots, dx_k)$ = the **vector difference** between the two points
and $ds := \|d\mathbf{r}\| = \sqrt{(dx_1)^2 + \dots + (dx_k)^2}$ = the **distance** between the two points
 $\approx$ the **arc length** of the bit of curve between the two points,
and we imagine them as being infinitely small.

Remark 47.2.

Let C be a C^1 curve given parametrically by $C = \{\mathbf{g}(t) = (x_1(t), \dots, x_k(t)) \in \mathbb{R}^k : a \leq t \leq b\}$.
Let $\mathbf{r} = \mathbf{g}(t)$ and $\mathbf{r} + d\mathbf{r} = \mathbf{g}(t + dt)$ be two nearby points.
Then $d\mathbf{r} = \mathbf{g}(t + dt) - \mathbf{g}(t) \approx \mathbf{g}'(t)dt := \left(\frac{dx_1(t)}{dt}, \dots, \frac{dx_k(t)}{dt}\right)dt$ and

$$\int_C d\mathbf{r} := \int_a^b \mathbf{g}'(t)dt = \mathbf{g}(b) - \mathbf{g}(a).$$

Moreover, $ds := \|d\mathbf{r}\| \approx \|\mathbf{g}'(t)\|dt = \sqrt{\left(\frac{dx_1(t)}{dt}\right)^2 + \dots + \left(\frac{dx_k(t)}{dt}\right)^2}dt$ and

$$\int_C ds := \int_a^b \|\mathbf{g}'(t)\|dt = \text{the total arc length of the curve } C.$$

Naturally, we take $\int_C ds$ as the definition of **arc length** for a smooth curve C .

Remark 47.3.

Let C be a C^1 curve given parametrically by $C = \{\mathbf{g}(t) \in \mathbb{R}^k : a \leq t \leq b\}$.
The parametrization $\mathbf{r} = \mathbf{g}(t)$ determines an **orientation** for C ,
that is, the “forward” direction and the “backward” direction
being the directions in which the point $\mathbf{g}(t)$ moves as t increases and decreases respectively.
The integral $\int_C d\mathbf{r}$ depends on the parametrization
but only switches the sign if the orientation is reversed.
In contrast, the arc length $\int_C ds$ is independent of parametrization and orientations.

Definition 47.4.

Let C be a **piecewise** C^1 curve in $\mathbb{R}^k$,
that is, it can be parameterized by $C = \{\mathbf{g}(t) \in \mathbb{R}^k : a \leq t \leq b\}$,
where $\mathbf{g} : [a, b] \rightarrow \mathbb{R}^k$ is a **piecewise** C^1 function on $[a, b]$,
i.e., $\mathbf{g}$ is continuous and injective on $[a, b]$ such that $\mathbf{g}'$ is well defined and is continuous on $[a, b]$
except perhaps at finitely many points t_i ’s, at which the one-sided limits $\lim_{t \rightarrow t_i^\pm} \mathbf{g}'(t)$ exist.
By a theorem given before, we have that $\|\mathbf{g}'(t)\|$ is Riemann integrable on $[a, b]$.
We define the **arc length** of C to be

$$\int_C ds := \int_a^b \|\mathbf{g}'(t)\|dt.$$

48 Line integrals

Definition 48.1.

Let C be a piecewise C^1 curve in $\mathbb{R}^k$ parametrized by $C = \{\mathbf{g}(t) \in \mathbb{R}^k : a \leq t \leq b\}$, let U be an open set such that $C \subseteq U \subseteq \mathbb{R}^k$, and $f : U \rightarrow \mathbb{R}$ be continuous on U .

We define the **line integral of the real-valued function f along C**

by $\int_C f ds = \int_a^b f(\mathbf{g}(t)) \|\mathbf{g}'(t)\| dt$ and the **average of f over C** by $\frac{\int_C f ds}{\int_C ds}$.

If $\mathbf{f} : U \rightarrow \mathbb{R}^m$ be continuous on $U \subseteq \mathbb{R}^k$ and has the components $\mathbf{f} = (f_1, \dots, f_m)$, then we define the **line integral of the vector-valued function $\mathbf{f}$ along C**

by $\int_C \mathbf{f} ds = (\int_C f_1 ds, \dots, \int_C f_m ds)$.

If $m = k$, we define the **line integral of the vector field $\mathbf{f}$ along C**

by $\int_C \mathbf{f} \cdot d\mathbf{r} = \int_a^b \mathbf{f}(\mathbf{g}(t)) \cdot \mathbf{g}'(t) dt$.

Remark 48.2.

1. If $f : U \rightarrow \mathbb{R}$ with $f \geq 0$,
then $\int_C f ds$ = the area of the region along C bounded above by f and below by the curve C
2. The integrals $\int_C f ds$ and $\int_C \mathbf{f} ds$ are independent of the parametrization and the orientation.
3. The integral $\int_C \mathbf{f} \cdot d\mathbf{r}$ is independent of the parametrization as long as the orientation is unchanged and it requires a factor of -1 if the orientation is reversed.
4. For the point $\mathbf{r} = \mathbf{g}(t)$ on C , define $\mathbf{u}(\mathbf{g}(t)) = \frac{\mathbf{g}'(t)}{\|\mathbf{g}'(t)\|}$ and $f_{\text{tang}}(\mathbf{g}(t)) = \mathbf{f}(\mathbf{g}(t)) \cdot \mathbf{u}(\mathbf{g}(t))$,
i.e., $\mathbf{u}(\mathbf{g}(t))$ is the unit tangent vector to C in the forward direction at the point $\mathbf{g}(t)$
and $f_{\text{tang}}(\mathbf{r})$ is the component of $\mathbf{f}(\mathbf{r})$ in the direction of $\mathbf{u}(\mathbf{r})$.
Then $\int_C \mathbf{f} \cdot d\mathbf{r} = \int_C f_{\text{tang}} ds$ and f_{tang} changes sign if the orientation is reversed.
5. If $\mathbf{f}$ is a force field, then $\int_C \mathbf{f} \cdot d\mathbf{r}$ represents the **work** done by the force $\mathbf{f}$ on a particle that traverses along the curve C .
6. Sometimes, we write $\int_C \mathbf{f} \cdot d\mathbf{r} = \int_C f_1 dx_1 + \dots + f_k dx_k = \int_a^b (f_1(\mathbf{g}(t))g'_1(t) + \dots + f_k(\mathbf{g}(t))g'_k(t)) dt$
and call the integrand in a **differential form**.
If $k = 1$, then $\int_C \mathbf{f} \cdot d\mathbf{r}$ depends on the orientation coming through C .

Proposition 48.3.

Let $\mathbf{f} : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be a continuous map and C be a C^1 curve in $\mathbb{R}^k$. Then

1. if $k = 1$ and $C = [a, b]$, then $\|\int_C \mathbf{f} ds\| = \|\int_{t=a}^b \mathbf{f}(t) dt\| \leq \int_C \|\mathbf{f}\| ds = \int_{t=a}^b \|\mathbf{f}(t)\| dt$;
2. $\|\int_C \mathbf{f} ds\| \leq \int_C \|\mathbf{f}\| ds = \int_a^b \|\mathbf{f}(\mathbf{g}(t))\| \|\mathbf{g}'(t)\| dt$, where $C = \{\mathbf{g}(t) \in \mathbb{R}^k : a \leq t \leq b\}$.
3. if $m = k$, then $|\int_C \mathbf{f} \cdot d\mathbf{r}| \leq \int_C \|\mathbf{f}\| ds$.

Theorem 48.4 (another definition of arc length).

Let $\mathbf{g} : [a, b] \rightarrow \mathbb{R}^k$ be continuous (not necessarily injective) on $[a, b]$.

For a partition $P = \{a = t_0 < \dots < t_n = b\}$ of $[a, b]$, define $\tilde{L}_g(P) = \sum_{i=1}^n \|\mathbf{g}(t_i) - \mathbf{g}(t_{i-1})\|$.

If $\mathbf{g}$ is C^1 on $[a, b]$, then $\mathbf{g}$ is **rectifiable** on $[a, b]$,

that is, the set $\{\tilde{L}_g(P) : P \text{ is a partition of } [a, b]\} \subseteq \mathbb{R}^1$ is bounded;

moreover, $\sup\{\tilde{L}_g(P) : P \text{ is a partition of } [a, b]\} = \int_a^b \|\mathbf{g}'(t)\| dt$.

One may use the above supremum as the definition of the arc length of a curve parametrized by $\mathbf{g}$,

even for the case when the parametrization is not injective or the $\mathbf{g}$ is not rectifiable,

provided that the supremum of a set not bounded above is defined to be ∞ .

The curve is an approximation of piecewise straight lines with end points on the curve.

Remark 48.5.

The limit of the arc lengths of curves may not be equal to the arc length of the limit curve,

that is, there exists a sequence of curves $\{\alpha_n : [0, 1] \rightarrow \mathbb{R}^k\}_{n=1}^\infty$ and a curve $\alpha : [0, 1] \rightarrow \mathbb{R}^k$

such that $\lim_{n \rightarrow \infty} \alpha_n(t) = \alpha(t)$ for all $t \in [0, 1]$ and $\lim_{n \rightarrow \infty} \int_{\alpha_n([0, 1])} ds \neq \int_{\alpha([0, 1])} ds$.

Consider the unit interval and the triangle curves, the circular curves, the Cantor function,

and the sine curves (a smooth one). One can show that the arc length of the Cantor function is TWO.

49 Green's theorem

Theorem 49.1 (Green's theorem or Stoke's theorem in the plane).

Let $D \subseteq \{(x, y, z) \in \mathbb{R}^3 : x, y \in \mathbb{R}, z = 0\}$ be a **regular region** in $\mathbb{R}^2$,

i.e., a compact set in $\mathbb{R}^2$ that is the closure of its interior.

Suppose that D has a piecewise smooth boundary ∂D and has the positive orientation,

where a **piecewise smooth boundary** ∂D means

∂D consists of a finite union of disjoint and piecewise smooth simply closed curves,

a **simply closed curve** means

that a curve begins and ends at the same point and has no intersection with itself,

the **positive orientation** means

traveling on ∂D in its positive direction keeps D to the left.

If $\mathbf{f} : D \rightarrow \mathbb{R}^3$ is C^1 on D with $\mathbf{f}(x, y, 0) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j} + 0\mathbf{k}$, then

$$\oint_{\partial D} \mathbf{f} \cdot d\mathbf{r} = \iint_D \text{curl}(\mathbf{f}) \cdot \mathbf{k} \, dA$$

or equivalently $\oint_{\partial D} [Pdx + Qdy] = \iint_D \left(\frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right) dA$,

where $\oint_{\partial D}$ denotes the sum of the line integrals around each of the closed curve in ∂D and

$$\text{curl}(\mathbf{f}) = \nabla \times \mathbf{f} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{bmatrix}$$

is called the **curl** of $\mathbf{f} = (P, Q, R)$.

Notice that here ∂D denotes the geometric boundary of D ,

which is equal to the topological boundary of D in $\mathbb{R}^2$, but not equal to the topological boundary of D in $\mathbb{R}^3$.

The FTC says that we can calculate an integral on an interval based on its antiderivative on the end points,

while Green's theorem says that we can calculate an integral over a planar region

based on a line integral of a vector field on the boundary of the region;

later, the high-dimensional Stoke's theorem will say that we can calculate an integral over a k -dimensional region based on an integral on the boundary of the region;

Corollary 49.2.

Let $D \subseteq \{(x, y, 0) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$ be a regular region in $\mathbb{R}^2$

with a piecewise smooth boundary ∂D ,

and let $\mathbf{n}(x, y, 0)$ be the unit normal vector in $\mathbb{R}^3$ to ∂D pointing out of D at $(x, y, 0) \in \partial D$

with $\mathbf{n}(x, y, 0) = n_1(x, y)\mathbf{i} + n_2(x, y)\mathbf{j} + 0\mathbf{k}$.

If $\mathbf{f} : D \rightarrow \mathbb{R}^3$ is C^1 on D with $\mathbf{f}(x, y, 0) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j} + 0\mathbf{k}$, then

$$\oint_{\partial D} \mathbf{f} \cdot \mathbf{n} \, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA.$$

Corollary 49.3.

Let $D \subseteq \{(x, y, 0) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$ be a regular region in $\mathbb{R}^2$

with a piecewise smooth boundary ∂D ,

and let $\mathbf{n}(x, y, 0)$ be the unit normal vector in $\mathbb{R}^3$ to ∂D pointing out of D at $(x, y, 0) \in \partial D$

with $\mathbf{n}(x, y, 0) = n_1(x, y)\mathbf{i} + n_2(x, y)\mathbf{j} + 0\mathbf{k}$.

Let $\mathbf{f} : D \rightarrow \mathbb{R}^3$ be C^1 on D with $\mathbf{f}(x, y, 0) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j} + 0\mathbf{k}$.

If $\mathbf{f}$ is **conservative** on D , that is, $\mathbf{f} = \nabla g$ for some C^2 real-valued function g on D , then

$$\oint_{\partial D} \mathbf{f} \cdot d\mathbf{r} = 0 \quad \text{and} \quad \oint_{\partial D} \mathbf{f} \cdot \mathbf{n} \, ds = \iint_D \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \right) dA.$$

Note that $\nabla g \cdot \mathbf{n}$ is the directional derivative of g in the unit normal direction to ∂D

pointing out of D and is called the **normal derivative** of g to ∂D ,

and $\Delta g := \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2}$ is called the **Laplacian** of g .

50 Surface area and surface integrals

Definition 50.1.

Let $D \subseteq \mathbb{R}^2$ be connected and open, $\mathbf{g}: D \rightarrow \mathbb{R}^3$ be C^1 and one-to-one on D , and let $E = \{\mathbf{g}(x, y) \in \mathbb{R}^3 : (x, y) \in D\}$ be a smooth surface in $\mathbb{R}^3$ parametrized by $\mathbf{g}$. For an infinitesimal rectangle $D_1 \subseteq D$ with vertices (x, y) , $(x + dx, y)$, $(x, y + dy)$, and $(x + dx, y + dy)$, we have $\mathbf{g}(x + dx, y) - \mathbf{g}(x, y) \approx \frac{\partial \mathbf{g}}{\partial x}(x, y)dx$ and $\mathbf{g}(x, y + dy) - \mathbf{g}(x, y) \approx \frac{\partial \mathbf{g}}{\partial y}(x, y)dy$; hence $\|\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}\|dxdy \approx$ the **area** of $\mathbf{g}(D_1)$ in E . From now on, we denote $dS = \|\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}\|dA \approx$ the **area** of $\mathbf{g}(D_1)$ in E . If D is a Jordan measurable set in $\mathbb{R}^2$, we define the **area** of E by $\iint_E dS = \iint_D \|\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}\|dA$.

Recall that writing $\mathbf{g} = (g_1, g_2, g_3)$, we have $\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial g_1}{\partial x} & \frac{\partial g_2}{\partial x} & \frac{\partial g_3}{\partial x} \\ \frac{\partial g_1}{\partial y} & \frac{\partial g_2}{\partial y} & \frac{\partial g_3}{\partial y} \end{bmatrix}$.

Remark 50.2.

1. The area $\iint_E dS$ is independent of the parametrization of E .
2. If the surface is the graph of a function $z = g(x, y)$, that is, the surface is parametrized by $\mathbf{g}(x, y) = (x, y, h(x, y))$ with $(x, y) \in D$, then $\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} = -\frac{\partial h}{\partial x}\mathbf{i} - \frac{\partial h}{\partial y}\mathbf{j} + \mathbf{k}$ and so $dS = \sqrt{(\frac{\partial h}{\partial x})^2 + (\frac{\partial h}{\partial y})^2 + 1} dA$.

Definition 50.3.

Let E be a surface in $\mathbb{R}^3$ parametrized by a one-to-one C^1 function $\mathbf{g}$ on a Jordan measurable set $D \subseteq \mathbb{R}^2$. Let $f: E \rightarrow \mathbb{R}^1$ be continuous on E . We define the **surface integral of the real-valued function f over E** by $\iint_E f dS = \iint_D f(\mathbf{g}(x, y))\|\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}\|dA$.

Let E be a **piecewise smooth surface**, that is, E is the union of finitely many pieces $E_1, \dots, E_n$ that satisfy the following:

1. Each E_i admits a smooth parametrization as discussed above;
2. The intersection $E_i \cap E_j$ are either empty or finitely unions of smooth curve.

We define the **surface integral of the real-valued function f over E** by $\iint_E f dS = \sum_{i=1}^n \iint_{E_i} f dS$.

Definition 50.4.

Let E be a smooth surface in $\mathbb{R}^3$ with a specified **orientation**, that is, a specification of which side of E is the “positive” side. Let $\mathbf{n}$ be the unit normal vector of E in $\mathbb{R}^3$ pointing along the positive side of E . Let $\mathbf{f}: E \rightarrow \mathbb{R}^3$ be continuous on E . We define the **surface integral of the vector field $\mathbf{f}$ over E** by $\iint_E \mathbf{f} \cdot \mathbf{n} dS$. If E is parametrized by $\mathbf{g}$ on $D \subseteq \mathbb{R}^2$, the orientation of E is specified by the unit normal vector $\mathbf{n}$ along the direction of $\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}$ or $-\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}$.

51 Surface area and surface integrals(continued)

Remark 51.1.

1. The integral $\iint_E \mathbf{f} \cdot \mathbf{n} \, dS$ is independent of the choice of parametrization as long as the parametrization induced the specified orientation of E ; switch to the opposite orientation results in multiplying the integral by -1 .
2. If E is a **non-orientable** surface such as the **Möbius band**, then $\iint_E \mathbf{f} \cdot \mathbf{n} \, dS$ is not defined.
3. If E is parametrized by $\mathbf{g}(x, y)$ with $(x, y) \in D$,
then $\mathbf{n} = \pm \frac{\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}}{\left\| \frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} \right\|}$ and $\iint_E \mathbf{f} \cdot \mathbf{n} \, dS = \pm \iint_D \mathbf{f}(\mathbf{g}(x, y)) \cdot \left(\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} \right) \, dA$.
4. If E is parametrized by $(x, y, h(x, y))$ with $(x, y) \in D$ and $\mathbf{f} = (f_1, f_2, f_3)$
then $\mathbf{n} = \pm \frac{-\frac{\partial h}{\partial x} \mathbf{i} - \frac{\partial h}{\partial y} \mathbf{j} + \mathbf{k}}{\sqrt{\left(\frac{\partial h}{\partial x}\right)^2 + \left(\frac{\partial h}{\partial y}\right)^2 + 1}}$ depending on the specified orientation of E
and $\iint_E \mathbf{f} \cdot \mathbf{n} \, dS = \pm \iint_D -f_1(x, y, h(x, y)) \frac{\partial h}{\partial x} - f_2(x, y, h(x, y)) \frac{\partial h}{\partial y} + f_3(x, y, h(x, y)) \, dA$.
5. Notice that $\mathbf{f} \cdot \mathbf{n}$ is the **normal component of $\mathbf{f}$ along E** ;
it is positive or negative according as $\mathbf{f}$ points into the positive or negative side of E .
If $\mathbf{f}$ represents the flow of some substance through the surface E :
the magnitude of $\mathbf{f}(\mathbf{g}(x, y))$ is the rate of flow of the substance past $\mathbf{g}(x, y)$
and its direction is the direction of the flow,
then the integral $\iint_E \mathbf{f} \cdot \mathbf{n} \, dS$ represents the **net flow**, or **flux**, of $\mathbf{f}$
across the surface E from the negative side to the positive side.

52 Vector derivatives

Definition 52.1.

Let ∇ be the k -tuple of partial differential operator: $\nabla = (\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_k})$.

Let $g: \mathbb{R}^k \rightarrow \mathbb{R}$ be C^1 .

The **gradient** of g is the vector-valued function defined by

$$\text{grad}(g) = \nabla g = (\frac{\partial g}{\partial x_1}, \dots, \frac{\partial g}{\partial x_k}).$$

Let $U \subseteq \mathbb{R}^k$ and $\mathbf{f}: U \rightarrow \mathbb{R}^k$ be C^1 with $\mathbf{f} = (f_1, \dots, f_k)$.

The **divergence** of $\mathbf{f}$ is the real-valued function defined by

$$\text{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \sum_{i=1}^k \frac{\partial f_i}{\partial x_i}.$$

If $k = 3$,

the **curl** of $\mathbf{f}$ is the vector-valued function defined by

$$\text{curl}(\mathbf{f}) = \nabla \times \mathbf{f} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ f_1 & f_2 & f_3 \end{bmatrix}.$$

Note that real-valued $\xrightarrow{\text{grad}}$ vector-valued $\xrightarrow{\text{curl}}$ vector-valued $\xrightarrow{\text{div}}$ real-valued.

Proposition 52.2.

Let $f, g: \mathbb{R}^3 \rightarrow \mathbb{R}$ be C^1 and let $\mathbf{F}, \mathbf{G}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be C^1 . Then

1. $\text{grad}(fg) = f \text{grad}(g) + g \text{grad}(f)$;
2. $\text{grad}(\mathbf{F} \cdot \mathbf{G}) = (\mathbf{F} \cdot \nabla) \mathbf{G} + \mathbf{F} \times \text{curl}(\mathbf{G}) + (\mathbf{G} \cdot \nabla) \mathbf{F} + \mathbf{G} \times \text{curl}(\mathbf{F})$;
3. $\text{curl}(f\mathbf{G}) = f \text{curl}(\mathbf{G}) + \text{grad}(f) \times \mathbf{G}$;
4. $\text{curl}(\mathbf{F} \times \mathbf{G}) = (\mathbf{G} \cdot \nabla) \mathbf{F} + \text{div}(\mathbf{G}) \mathbf{F} - (\mathbf{F} \cdot \nabla) \mathbf{G} - \text{div}(\mathbf{F}) \mathbf{G}$;
5. $\text{div}(f\mathbf{G}) = f \text{div}(\mathbf{G}) + \text{grad}(f) \cdot \mathbf{G}$;
6. $\text{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \text{curl}(\mathbf{F}) - \mathbf{F} \cdot \text{curl}(\mathbf{G})$.

Here $\mathbf{F} \cdot \nabla = (F_1, F_2, F_3) \cdot (\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3}) = \sum_{i=1}^3 F_i \frac{\partial}{\partial x_i}$.

Proposition 52.3.

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be C^2 and let $\mathbf{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be C^2 . Then

1. $\text{curl}(\text{grad}(f)) = \mathbf{0}$;
2. $\text{div}(\text{curl}(\mathbf{F})) = 0$;
3. $\text{div}(\text{grad}(f)) = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \frac{\partial^2 f}{\partial x_3^2}$,
called the **Laplacian of the real-valued function** f and denoted by $\nabla^2 f$ or Δf ;
4. $\text{grad}(\text{div}(\mathbf{F})) - \text{curl}(\text{curl}(\mathbf{F})) = \nabla^2 F_1 \mathbf{i} + \nabla^2 F_2 \mathbf{j} + \nabla^2 F_3 \mathbf{k}$,
called the **Laplacian of the vector-valued** $\mathbf{F}$ and denoted by $\nabla^2 \mathbf{F}$ or $\Delta \mathbf{F}$.

53 Gauss's divergence theorem and Green's formula

Theorem 53.1 (Gauss's divergence theorem).

Let E be a (solid) **regular region** in $\mathbb{R}^3$, i.e., a compact set in $\mathbb{R}^3$ which is the closure of its interior, with piecewise smooth boundary ∂E which is an oriented surface so that the positive unit normal vector $\mathbf{n}$ to ∂E pointing out of E .

Let $\mathbf{f} : E \rightarrow \mathbb{R}^3$ be C^1 on E . Then

$$\iint_{\partial E} \mathbf{f} \cdot \mathbf{n} \, dS = \iiint_E \operatorname{div}(\mathbf{f}) \, dV.$$

Remark 53.2.

Let $E_\epsilon = \overline{B}_\epsilon(\mathbf{a})$, the closed ball with center at $\mathbf{a} \in \mathbb{R}^3$ and radius $\epsilon > 0$ small.

Let $\mathbf{f} : E_\epsilon \rightarrow \mathbb{R}^3$ be C^1 .

If $\epsilon > 0$ is very small, then $\mathbf{f}(\mathbf{r}) \approx \mathbf{f}(\mathbf{a})$ for all $\mathbf{r} \in E_\epsilon$.

By Gauss's divergence theorem, we have $\operatorname{div}(\mathbf{f}(\mathbf{a})) = \lim_{\epsilon \rightarrow 0} \frac{3}{4\pi\epsilon^3} \iint_{\partial E_\epsilon} \mathbf{f} \cdot \mathbf{n} \, dS$.

Let $\mathbf{f}$ represent the flow of some substance through E_ϵ .

Then the above integral is the flux of $\mathbf{f}$ across ∂E_ϵ from the inside to the outside of the ball, that is, the amount of substance flowing out E_ϵ minus the amount of substance flowing in; thus the condition $\operatorname{div}(\mathbf{f})$ means that there is a **net** outflow near $\mathbf{a}$, that is, $\mathbf{f}$ tends to **diverge** from $\mathbf{a}$.

Corollary 53.3 (Green's formula).

Let E be a **regular region** in $\mathbb{R}^3$

with piecewise smooth boundary ∂E which is oriented

so that the positive unit normal vector $\mathbf{n}$ to ∂E pointing out of E .

Let $f, g : E \rightarrow \mathbb{R}$ be C^2 on E . Then

$$\begin{aligned} \iint_{\partial E} f \nabla g \cdot \mathbf{n} \, dS &= \iiint_E (\nabla f \cdot \nabla g + f \nabla^2 g) \, dV, \\ \iint_{\partial E} (f \nabla g - g \nabla f) \cdot \mathbf{n} \, dS &= \iiint_E (f \nabla^2 g - g \nabla^2 f) \, dV. \end{aligned}$$

Note that $\nabla g \cdot \mathbf{n}$ is the directional derivative of g in the outward normal direction to ∂E and is called the **normal derivative** of g to ∂E ; it is also denoted by $\frac{\partial g}{\partial n}$ in the textbook.

54 Stokes' theorem

Theorem 54.1 (Stokes' theorem).

Let D be an oriented surface in $\mathbb{R}^3$

that is completely enclosed by a piecewise smooth simply closed curve ∂D

(here ∂D is the geometric boundary of the manifold D)

and has a continuously varying unit normal vector $\mathbf{n}$

such that ∂D is **oriented consistently** with $\mathbf{n}$,

i.e., if you walk around ∂D in the positive direction of ∂D

with your head in the positive direction $\mathbf{n}$ of D ,

then the surface D is always on your left-hand side;

(equivalently, if at any point $\mathbf{r} \in \partial D$,

$\mathbf{t}$ is the unit tangent vector to ∂D in the positive direction of ∂D

and $\mathbf{n}$ is the unit normal vector to D in the positive direction of D ,

then $\mathbf{n} \times \mathbf{t}$, considered as an arrow emanating from $\mathbf{r}$, points into D).

Let U be an open neighborhood of D in $\mathbb{R}^3$ and let $\mathbf{f} : U \rightarrow \mathbb{R}^3$ be C^1 .

Then

$$\oint_{\partial D} \mathbf{f} \cdot d\mathbf{r} = \iint_D \text{curl}(\mathbf{f}) \cdot \mathbf{n} \, dS.$$

Remark 54.2.

1. If D is a planar disk in $\mathbb{R}^3$, then Stokes' theorem derives Green's theorem.

2. Every simply closed curve in $\mathbb{R}^3$ is the boundary of infinitely many surfaces in $\mathbb{R}^3$.

Stokes' theorem says that

if ∂D is a simply closed curve in $\mathbb{R}^3$ and D is any oriented surface bounded by ∂D ,

then $\oint_{\partial D} \mathbf{f} \cdot d\mathbf{r} = \iint_D \text{curl}(\mathbf{f}) \cdot \mathbf{n} \, dS$.

Remark 54.3.

Let $\mathbf{n}$ be a unit vector in $\mathbb{R}^3$ and $\mathbf{a}$ be a point in $\mathbb{R}^3$.

Let D_ϵ be a two-dimensional disk with center at $\mathbf{a} \in \mathbb{R}^3$ and radius $\epsilon > 0$ small

in the plane perpendicular to $\mathbf{n}$, oriented so that $\mathbf{n}$ is the positive normal to D_ϵ .

Let $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be C^1 .

If $\epsilon > 0$ is very small, then $\mathbf{f}(\mathbf{r}) \approx \mathbf{f}(\mathbf{a})$ for all $\mathbf{r} \in D_\epsilon$.

Let ∂D_ϵ be the circle with center $\mathbf{a}$ and radius ϵ in the plane perpendicular to $\mathbf{n}$,

traversed counterclockwise as heading toward the direction of $\mathbf{n}$.

By Stokes' theorem, we have $\text{curl}(\mathbf{f}(\mathbf{a})) \cdot \mathbf{n} = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi \epsilon^2} \oint_{\partial D_\epsilon} \mathbf{f} \cdot d\mathbf{r}$.

Let $\mathbf{f}$ represent a force field in $\mathbb{R}^3$.

Then $\oint_{\partial D_\epsilon} \mathbf{f} \cdot d\mathbf{r}$ is the work done by $\mathbf{f}$ on a particle that moves around ∂D_ϵ .

Thus $\text{curl}(\mathbf{f}(\mathbf{a})) \cdot \mathbf{n}$ represents the tendency of the force $\mathbf{f}$ to push the particle around ∂D_ϵ ,

counterclockwise if it is positive and clockwise if it is negative as viewed from the $\mathbf{n}$ -side.

Corollary 54.4.

Let D be a **closed surface** (i.e., a surface with no boundary) in $\mathbb{R}^3$

with unit outward normal $\mathbf{n}$, and let $\mathbf{f} : D \rightarrow \mathbb{R}^3$ be C^1 .

Then

$$\iint_D \text{curl}(\mathbf{f}) \cdot \mathbf{n} \, dS = 0.$$

55 High-dimensional Gauss's divergence theorem

Recall 55.1 (Gauss's divergence theorem).

Let E be a (solid) **regular region** in $\mathbb{R}^3$, i.e., a compact set in $\mathbb{R}^3$ which is the closure of its interior, with piecewise smooth boundary ∂E which is an oriented surface so that the positive unit normal vector $\mathbf{n}$ to ∂E pointing out of E .

Let $\mathbf{f} : E \rightarrow \mathbb{R}^3$ be C^1 on E . Then

$$\iint_{\partial E} \mathbf{f} \cdot \mathbf{n} dS = \iiint_E \operatorname{div}(\mathbf{f}) dV.$$

Recall the following:

1. If ∂E is parameterized by $\mathbf{g}(x, y)$ with $(x, y) \in D$,

$$\text{then } \mathbf{n} = \frac{\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y}}{\left\| \frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} \right\|} \text{ and } \mathbf{n} dS = \left(\frac{\partial \mathbf{g}}{\partial x} \times \frac{\partial \mathbf{g}}{\partial y} \right) dA = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial g_1}{\partial x} & \frac{\partial g_2}{\partial x} & \frac{\partial g_3}{\partial x} \\ \frac{\partial g_1}{\partial y} & \frac{\partial g_2}{\partial y} & \frac{\partial g_3}{\partial y} \end{bmatrix} dx dy.$$

2. The divergence of $\mathbf{f}$ is the real-valued function

$$\text{defined by } \operatorname{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_k} \right) \cdot (f_1, \dots, f_k) = \sum_{i=1}^k \frac{\partial f_i}{\partial x_i}.$$

Theorem 55.2 (high-dimensional Gauss's divergence theorem).

Let E be a (solid) **regular region** in $\mathbb{R}^k$, i.e., a compact set in $\mathbb{R}^k$ which is the closure of its interior, with piecewise smooth boundary ∂E which is an oriented **hypersurface** so that the positive unit normal vector $\mathbf{n}$ to ∂E pointing out of E .

Let $\mathbf{f} : E \rightarrow \mathbb{R}^k$ be C^1 on E . Then

$$\int \cdots \int_{\partial E} \mathbf{f} \cdot \mathbf{n} dS^{k-1} = \iiint_E \operatorname{div}(\mathbf{f}) dV^k,$$

where, if ∂E is parameterized by $\mathbf{g}(x_1, \dots, x_{k-1})$ with $(x_1, \dots, x_{k-1}) \in D$,

$$\text{then } \mathbf{n} dS^{k-1} = \det \begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_k \\ \partial_1 g_1 & \cdots & \partial_1 g_k \\ \vdots & & \vdots \\ \partial_{k-1} g_1 & \cdots & \partial_{k-1} g_k \end{bmatrix} dx_1 \cdots dx_{k-1},$$

where $\{\mathbf{e}_1, \dots, \mathbf{e}_k\}$ is the standard orthonormal basis on $\mathbb{R}^k$.

56 Differential forms and the general Stokes' theorem

Recall 56.1 (Stokes' theorem).

Let M be an oriented surface in $\mathbb{R}^3$ that is completely enclosed by a piecewise smooth simply closed curve ∂M (here ∂M is a manifold boundary of M) and has a continuously varying unit normal vector $\mathbf{n}$ such that ∂M is **oriented consistently** with $\mathbf{n}$.

Let U be an open neighborhood of M in $\mathbb{R}^3$ and let $\mathbf{f}: U \rightarrow \mathbb{R}^3$ be C^1 .

Then

$$\oint_{\partial M} \mathbf{f} \cdot d\mathbf{r} = \iint_M \text{curl}(\mathbf{f}) \cdot \mathbf{n} dS.$$

Remark 56.2.

1. If $\{x_1, x_2, x_3\}$ is the stand basis on $\mathbb{R}^3$, ∂M is parameterized by $t \mapsto \mathbf{g}(t)$, and M is parameterized by $(u_1, u_2) \mapsto \mathbf{h}(u_1, u_2)$, then

$$\int_{\mathbf{g}^{-1}(\partial M)} \sum_{i=1}^3 f_i(\mathbf{g}(t)) \frac{dg_i(t)}{dt} dt = \iint_{\mathbf{h}^{-1}(M)} \det \begin{bmatrix} \frac{\mathbf{i}}{\partial x_1} & \frac{\mathbf{j}}{\partial x_2} & \frac{\mathbf{k}}{\partial x_3} \\ f_1 & f_2 & f_3 \end{bmatrix} (\mathbf{h}(u_1, u_2)) \cdot \frac{\partial \mathbf{h}}{\partial u_1} \times \frac{\partial \mathbf{h}}{\partial u_2} du_1 du_2.$$

2. Let the 1-**form** $\omega = \sum_{i=1}^3 f_i(x_1, x_2, x_3) dx_i$, the 2-**form** $d\omega = \sum_{1 \leq i < j \leq 3} \left(\frac{\partial f_j}{\partial x_i} - \frac{\partial f_i}{\partial x_j} \right) dx_i \wedge dx_j$,

where d means the **exterior derivative**,

the **pullback** $\mathbf{g}^* \omega = \sum_{i=1}^3 f_i(\mathbf{g}(t)) \frac{dg_i(t)}{dt} dt$, where $t \mapsto \mathbf{g}(t) = (g_1(t), g_2(t), g_3(t))$,

and the **pullback** $\mathbf{h}^* d\omega = \sum_{1 \leq i < j \leq 3} \left(\frac{\partial f_j}{\partial x_i} - \frac{\partial f_i}{\partial x_j} \right) (\mathbf{h}(u_1, u_2)) \frac{\partial(h_i, h_j)}{\partial(u_1, u_2)} du_1 du_2$,

where $(u_1, u_2) \mapsto \mathbf{h}(u_1, u_2) = (h_1, h_2, h_3)$, and $\frac{\partial(h_i, h_j)}{\partial(u_1, u_2)}$ denotes the Jacobian of the maps $(u_1, u_2) \mapsto (h_i, h_j)$.

Then

$$\int_{\partial M} \omega \stackrel{\text{by line integral}}{=} \int_{\mathbf{g}^{-1}(\partial M)} \mathbf{g}^* \omega \stackrel{\text{by item 1}}{=} \iint_{\mathbf{h}^{-1}(M)} \mathbf{h}^* d\omega \stackrel{\text{by surface integral}}{=} \iint_M d\omega,$$

which indicate the integral of 1-form ω over the boundary ∂M equal to the integral of 2-form $d\omega$ over the surface M .

Definition 56.3.

On the high-dimensional space $\mathbb{R}^n$, we define the followings.

1. A 0-**form** is a function of the coordinates.
The basic 1-forms are the differentials of the coordinates: $dx_1, \dots, dx_n$.
A **general 1-form** is a linear combinations of these differentials $\sum_{i=1}^n f_i dx_i$, where each f_i is a functions of the coordinates.
2. The basic 2-forms are expressions $dx_i \wedge dx_j$ with $i < j$, and a **general 2-form** is a linear combinations of these.
3. The **wedge product** $\wedge$ involves orientation: $dx_i \wedge dx_j = -dx_j \wedge dx_i$ and $dx_i \wedge dx_i = 0$
4. Let ω be a k -form. Then it is a sum of terms of the form $f\beta$, where f is a function and β is a basic k -form.
The **exterior derivative** $d\omega$ of the k -form ω is defined to be the $(k+1)$ -form obtained by replacing each such term $f\beta$ by $df \wedge \beta$, where the differential $df = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j$.

5. Let $k \leq m \leq n$, and let $\{v_1, \dots, v_m\}$ and $\{x_1, \dots, x_n\}$ be the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively.

Let ω be a k -form on $\mathbb{R}^n$, say $\omega = \sum_{i_1 < \dots < i_k} a_{i_1 \dots i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}$.

Let $\mathbf{g}: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a smooth one-to-one function with $\mathbf{g} = (g_1, \dots, g_n)$.

The **pullback** of ω by $\mathbf{g}$ is defined to be the following k -form on $\mathbb{R}^m$

$$\mathbf{g}^* \omega = \sum_{i_1 < \dots < i_k} (a_{i_1 \dots i_k} \circ \mathbf{g}) dg_{i_1} \wedge \dots \wedge dg_{i_k} = \sum_{i_1 < \dots < i_k} \sum_{j_1 < \dots < j_k} (a_{i_1 \dots i_k} \circ \mathbf{g}) \frac{\partial(g_{i_1}, \dots, g_{i_k})}{\partial(v_{j_1}, \dots, v_{j_k})} dv_{j_1} \dots dv_{j_k},$$

where $\frac{\partial(g_{i_1}, \dots, g_{i_k})}{\partial(v_{j_1}, \dots, v_{j_k})}$ denotes the Jacobian of the function $(v_{j_1}, \dots, v_{j_k}) \mapsto (g_{i_1}, \dots, g_{i_k})$.

Let $m = k$, $\mathbf{g} = (g_1, \dots, g_n)$, and M be a k -dimensional manifold in $\mathbb{R}^n$ parameterized by $\mathbf{g}$ on $\mathbf{g}^{-1}(M)$ in $\mathbb{R}^k$.

The **integral of the k -form ω over M** is defined to be

$$\int \dots \int_M \omega = \int \dots \int_{\mathbf{g}^{-1}(M)} \mathbf{g}^* \omega = \int \dots \int_{\mathbf{g}^{-1}(M)} \sum_{i_1 < \dots < i_k} (a_{i_1 \dots i_k} \circ \mathbf{g}) \frac{\partial(g_{i_1}, \dots, g_{i_k})}{\partial(v_1, \dots, v_k)} dv_1 \dots dv_k.$$

Theorem 56.4. (the general Stokes' theorem)

Let M be a smooth, oriented k -dimensional submanifold of $\mathbb{R}^n$ with a piecewise smooth boundary ∂M ,

and let ∂M carry the orientation that is (in a suitable sense) compatible with the one on M .

If ω is a $(k-1)$ -form of class C^1 on an open set containing M ,

then $\int \dots \int_{\partial M} \omega = \iint \dots \int_M d\omega$.

57 Infinite series

Definition 57.1.

Let $\{a_i\}_{i=0}^{\infty}$ be a sequence of real numbers.

The notation $a_0 + a_1 + a_2 + \cdots + a_i + \cdots$, denoted by $\sum_{i=0}^{\infty} a_i$, is called an **infinite series**.

The number a_i is called the **i -th term** of the infinite series $\sum_{i=0}^{\infty} a_i$.

For $n \in \mathbb{N} \cup \{0\}$, define $S_n = \sum_{i=0}^n a_i$ to be the **n -th partial sum** of the infinite series $\sum_{i=0}^{\infty} a_i$.

The sequence $\{S_n\}_{n=0}^{\infty}$ is called the **sequence of partial sum** of the infinite series $\sum_{i=0}^{\infty} a_i$.

If $\lim_{n \rightarrow \infty} S_n$ exists as a finite number, we say that the series $\sum_{i=0}^{\infty} a_i$ is **convergent**

and write $\sum_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} S_n$ and call it the **sum** of the infinite series.

Otherwise, we say the infinite series is **divergent**;

in the case when $S_n \rightarrow \pm\infty$ as $n \rightarrow \infty$, we say that $\sum_{i=0}^{\infty} a_i = \pm\infty$.

Remark 57.2.

For any $k \in \mathbb{N}$, $\sum_{i=0}^{\infty} a_i$ and $\sum_{i=k}^{\infty} a_i$ are convergent or divergent simultaneously.

Proposition 57.3.

1. The series $\sum_{i=1}^{\infty} \frac{1}{i}$, called the **harmonic series**, is divergent.
2. For $a \neq 0$, the series $\sum_{i=0}^{\infty} ar^i$, called the **geometric series**,

$$\begin{cases} \text{has the partial sum } \sum_{i=0}^n ar^i = \frac{a(1-r^{n+1})}{1-r}, & \text{if } r \neq 1, \\ \text{is convergent and has the sum } \sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}, & \text{if } |r| < 1, \\ \text{is divergent,} & \text{if } |r| \geq 1. \end{cases}$$
3. Cauchy condensation test...

Theorem 57.4.

1. (Cauchy condition for convergence series) $\sum_{n=0}^{\infty} a_n$ is convergent
 $\iff$ for $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that if $m \geq N$ and $p \geq 0$ then $|\sum_{i=m}^{m+p} a_i| < \varepsilon$.
2. (divergence test) If $\sum_{i=0}^{\infty} a_i$ is convergent, then $\lim_{i \rightarrow \infty} a_i = 0$
3. Let $\sum_{i=0}^{\infty} a_i$ be a **telescoping series**,
i.e., there exists a sequence $\{b_i\}_{i=0}^{\infty}$ in $\mathbb{R}$ such that $a_0 = b_0$ and $a_i = b_i - b_{i-1}$ for all $i \geq 1$.
 Then $\sum_{i=0}^{\infty} a_i$ is convergent $\iff \lim_{i \rightarrow \infty} b_i$; in this case, $\sum_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} b_n$.

Theorem 57.5 (linearity of convergent series).

If $\sum_{i=0}^{\infty} a_i$ and $\sum_{i=0}^{\infty} b_i$ are convergent, then

1. $\sum_{i=0}^{\infty} (a_i + b_i)$ is convergent and $\sum_{i=0}^{\infty} (a_i + b_i) = \sum_{i=0}^{\infty} a_i + \sum_{i=0}^{\infty} b_i$.
2. For any $c \in \mathbb{R}$, $\sum_{i=0}^{\infty} ca_i$ is convergent and $\sum_{i=0}^{\infty} ca_i = c \sum_{i=0}^{\infty} a_i$.
3. $\sum_{i=0}^{\infty} (a_i - b_i)$ is convergent and $\sum_{i=0}^{\infty} (a_i - b_i) = \sum_{i=0}^{\infty} a_i - \sum_{i=0}^{\infty} b_i$.

Definition 57.6.

Let $\{a_i\}_{i=0}^{\infty}$ be a sequence of real numbers.

The product $a_0 a_1 a_2 \cdots a_i \cdots$, denoted by $\prod_{i=0}^{\infty} a_i$, is called an **infinite product**.

For $n \in \mathbb{N} \cup \{0\}$, define $P_n = \prod_{i=0}^n a_i = a_0 a_1 \cdots a_n$ to be

the **n -th partial product** of the infinite product $\prod_{i=0}^{\infty} a_i$.

The sequence $\{P_n\}_{n=0}^{\infty}$ is called the **sequence of partial product** of $\prod_{i=0}^{\infty} a_i$.

If $\lim_{n \rightarrow \infty} P_n$ exists as a finite number, we say that $\prod_{i=0}^{\infty} a_i$ is **convergent**

and write $\prod_{i=0}^{\infty} a_i = \lim_{n \rightarrow \infty} P_n$ and call it the **product** of the infinite product.

Otherwise, we say the infinite product is **divergent**;

in the case when $P_n \rightarrow \pm\infty$ as $n \rightarrow \infty$, we write $\prod_{i=0}^{\infty} a_i = \pm\infty$.

58 Taylor series

Recall 58.1 (from Theorems 22.1-22.3).

If $f : (-c, c) \rightarrow \mathbb{R}$ is C^∞ on $(-c, c)$,
then for any $m \in \mathbb{N}$, we have for $x \in (-c, c)$,

$$f(x) = \sum_{n=0}^m \frac{f^{(n)}(0)}{n!} x^n + R_{f,0,m}(x),$$

where $R_{f,0,m}(x)$ is the m th-order Taylor remainder for f and has different forms;

refer to Theorem 22.3 for $R_{f,0,m}(x) = \frac{f^{(m+1)}(z)}{(m+1)!} x^{m+1}$ with some z between x and 0 .

If $\lim_{m \rightarrow \infty} R_{f,0,m}(x) = 0$ for $x \in (-c, c)$, then the series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ for $x \in (-c, c)$,
called the **Taylor series** of f about the origin or the **Maclaurin series** of f .

Theorem 58.2.

Let $f : (-c, c) \rightarrow \mathbb{R}$ be C^∞ on $(-c, c)$, where $0 < c \leq \infty$.

1. If there exist $a > 0$ and $b > 0$ such that $|f^{(k)}(x)| \leq ab^k k!$ for all $x \in (-c, c)$ and $k \geq 0$,

then $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ for $|x| < \min(c, b^{-1})$.

2. If there exist $A > 0$ and $B > 0$ such that $|f^{(k)}(x)| \leq AB^k$ for all $x \in (-c, c)$ and $k \geq 0$,

then $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ for $|x| < c$.

Proposition 58.3.

1. $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ for all $x \in \mathbb{R}$.

2. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all $x \in \mathbb{R}$.

3. $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ for all $x \in (-1, 1]$.

59 Series with nonnegative terms I: the integral test and the comparison tests

Theorem 59.1 (integral test).

Let $f : [1, \infty) \rightarrow (0, \infty)$ be a continuous and decreasing function,
and let $a_i = f(i)$ for all $i \in \mathbb{N}$.

Then

1. if $k, m \in \mathbb{N}$ with $1 \leq k < m$, then $\sum_{i=k+1}^m a_i \leq \int_k^m f(x)dx \leq \sum_{i=k}^{m-1} a_i$;
2. the series $\sum_{i=1}^{\infty} a_i$ is convergent $\iff$ the improper integral $\int_1^{\infty} f(x)dx$ is convergent;
3. if $\int_1^{\infty} f(x)dx$ is convergent,
then $\sum_{i=1}^{k-1} a_i + \int_k^{\infty} f(x)dx \leq \sum_{i=1}^{\infty} a_i \leq \sum_{i=1}^{k-1} a_i + \int_k^{\infty} f(x)dx + f(k)$ for all integer $k \geq 2$.

Theorem 59.2.

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, called the **p-series** or the **hyperharmonic series**,
is convergent if $p > 1$ and divergent if $p \leq 1$.

Theorem 59.3 (comparison test).

Let $\{a_i\}_{i=0}^{\infty}$ and $\{b_i\}_{i=0}^{\infty}$ be two sequences of real numbers.

Suppose that there exist $c \in \mathbb{R}$ and $m \in \mathbb{N}$ such that $0 \leq a_i \leq cb_i$ for all $i \geq m$.

Then

1. if $\sum_{i=0}^{\infty} b_i$ is convergent then $\sum_{i=0}^{\infty} a_i$ is convergent;
2. if $\sum_{i=0}^{\infty} a_i$ is divergent then $\sum_{i=0}^{\infty} b_i$ is divergent.

Theorem 59.4 (limit comparison test).

Let $\{a_i\}_{i=0}^{\infty}$ and $\{b_i\}_{i=0}^{\infty}$ be two sequences of positive real numbers.

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ exists as a finite number in $(0, \infty)$,
then $\sum_{i=0}^{\infty} a_i$ and $\sum_{i=0}^{\infty} b_i$ are either both convergent or both divergent.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum_{i=0}^{\infty} b_i$ is convergent, then $\sum_{i=0}^{\infty} a_i$ is convergent.
Hence, if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum_{i=0}^{\infty} a_i$ is divergent, then $\sum_{i=0}^{\infty} b_i$ is divergent.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum_{i=0}^{\infty} b_i$ is divergent, then $\sum_{i=0}^{\infty} a_i$ is divergent.
Hence, if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum_{i=0}^{\infty} a_i$ is convergent, then $\sum_{i=0}^{\infty} b_i$ is convergent.

60 Series with nonnegative terms II: the ratio and root tests

Theorem 60.1 (ratio test).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers.

1. If there exist $r < 1$, $m \in \mathbb{N}$ such that $\frac{a_{n+1}}{a_n} < r$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is convergent.
2. If there exists $m \in \mathbb{N}$ such that $\frac{a_{n+1}}{a_n} \geq 1$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is divergent.
3. Suppose $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$. Then
 - (a) if $L < 1$ then $\sum_{n=0}^{\infty} a_n$ is convergent;
 - (b) if $L > 1$ or $L = \infty$ then $\sum_{n=0}^{\infty} a_n$ is divergent;
 - (c) if $L = 1$ then the test fails.

Theorem 60.2 (root test).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers.

1. If there exist $r < 1$ and $m \in \mathbb{N}$ such that $a_n^{\frac{1}{n}} < r$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is convergent.
2. If there exists $m \in \mathbb{N}$ such that $a_n^{\frac{1}{n}} \geq 1$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is divergent.
3. Suppose $\lim_{n \rightarrow \infty} a_n^{\frac{1}{n}} = L$ or $\limsup_{n \rightarrow \infty} a_n^{1/n} = L$. Then
 - (a) if $L < 1$ then $\sum_{n=0}^{\infty} a_n$ is convergent;
 - (b) if $L > 1$ or $L = \infty$ then $\sum_{n=0}^{\infty} a_n$ is divergent;
 - (c) if $L = 1$ then the test fails.

Proposition 60.3.

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers.

1. $\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \leq \limsup_{n \rightarrow \infty} a_n^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$.
2. If $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$ exists, then $\lim_{n \rightarrow \infty} a_n^{\frac{1}{n}}$ exists and $\lim_{n \rightarrow \infty} a_n^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$.
3. There is an example such that $\lim_{n \rightarrow \infty} a_n^{\frac{1}{n}}$ exists but $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$ does not exist, e.g. a_n is $2^{-n/2}$ if n is even and is $2^{-(n-1)/2}$ if n is odd.

Thus, the root test, in theory, is more powerful than the ratio test.

However, the ratio test, in practice, is more useful than the root test,

especially when dealing with series whose terms involving factorials or similar sorts of products.

61 Series with nonnegative terms III: Raabe's test

Theorem 61.1 (Generalized version of Raabe's test).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers.

1. If there exist a real number $L > 1$ and an integer $N > 0$ such that $n(1 - \frac{a_{n+1}}{a_n}) \geq L$ for all $n \geq N$, then $\sum_{n=0}^{\infty} a_n$ is convergent.
2. If there exists an integer $N > 0$ such that $n(1 - \frac{a_{n+1}}{a_n}) \leq \frac{n}{n+1}$ for all $n \geq N$, then $\sum_{n=0}^{\infty} a_n$ is divergent.

Theorem 61.2 (Raabe's test).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of positive real numbers

such that $\lim_{n \rightarrow \infty} n(1 - \frac{a_{n+1}}{a_n}) = L$ (including the case when $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$).

1. If $L > 1$ or $L = \infty$ then $\sum_{n=0}^{\infty} a_n$ is convergent.
2. If $L < 1$ then $\sum_{n=0}^{\infty} a_n$ is divergent.
3. If $L = 1$ then the test fails.

Remark 61.3.

How to determine whether $\sum_{n=0}^{\infty} a_n$ is convergent or divergence?

1. Try the divergence test to decide divergence;
for instance, $1 - \frac{1}{2} + 1 - \frac{1}{3} + 1 - \frac{1}{4} + \dots$
2. If a_n is an algebraic function of n ,
say a rational function of n or a similar expression involving fractional powers of n ,
then try the comparison test by comparing with $\sum_{n=1}^{\infty} \frac{1}{n^p}$ for some p ;
for instance, $\sum_{n=0}^{\infty} \frac{\sqrt{n+1}}{n^2-4n+5}$.
3. If a_n involves expression with n is the exponent, try the ratio test or the root test;
for instance, $\sum_{n=0}^{\infty} \frac{n^2}{2^n}$.
4. If a_n involves factorial-like products, try the ratio test; for instance $\sum_{n=1}^{\infty} \frac{n!}{10^n}$
5. If the ratio test fails because $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$, try Raabe's test;
for instance, $\sum_{n=1}^{\infty} \frac{1 \cdot 4 \cdot 7 \cdots (3n+1)}{n^2 3^n n!}$.
6. The integral test may be used when you need to estimate numerical approximation
or when the series is near the borderline between convergence and divergence;
for instance, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ with p near 1.

62 Absolute and conditional convergence

Definition 62.1.

1. A series $\sum_{n=0}^{\infty} a_n$ is said to be **absolutely convergent** if the series $\sum_{n=0}^{\infty} |a_n|$ is convergent.
2. A series $\sum_{n=0}^{\infty} a_n$ is said to be **conditionally convergent** if $\sum_{n=0}^{\infty} a_n$ is convergent but $\sum_{n=0}^{\infty} |a_n|$ is divergent.

Theorem 62.2.

If $\sum_{n=0}^{\infty} a_n$ is absolutely convergent then $\sum_{n=0}^{\infty} a_n$ is convergent.

Theorem 62.3.

1. If $\sum_{n=0}^{\infty} a_n$ is absolutely convergent, then $\sum_{n=0}^{\infty} \max\{a_n, 0\}$ and $\sum_{n=0}^{\infty} \max\{-a_n, 0\}$ are both convergent.
2. If $\sum_{n=0}^{\infty} a_n$ is conditionally convergent, then $\sum_{n=0}^{\infty} \max\{a_n, 0\}$ and $\sum_{n=0}^{\infty} \max\{-a_n, 0\}$ are both divergent.

Theorem 62.4.

If $\sum_{n=0}^{\infty} a_n$ is absolutely convergent with sum S , then for any one-to-one and onto map $\sigma : \mathbb{N} \cup \{0\} \rightarrow \mathbb{N} \cup \{0\}$, one has that $\sum_{n=0}^{\infty} a_{\sigma(n)}$, called a **rearrangement** of $\sum_{n=0}^{\infty} a_n$, is also absolutely convergent with sum S .

Theorem 62.5.

If $\sum_{n=0}^{\infty} a_n$ is conditionally convergent, then the following holds.

1. For any $S \in \mathbb{R}$, there exists a one-to-one and onto map $\sigma : \mathbb{N} \cup \{0\} \rightarrow \mathbb{N} \cup \{0\}$ such that $\sum_{n=0}^{\infty} a_{\sigma(n)}$ is convergent with sum S .
2. There exists a one-to-one and onto map $\sigma : \mathbb{N} \cup \{0\} \rightarrow \mathbb{N} \cup \{0\}$ such that $\sum_{n=0}^{\infty} a_{\sigma(n)}$ is divergent.

Theorem 62.6 (comparison test for absolute convergence).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers. Then

1. if there exist $C > 0$, $\epsilon > 0$ and $m \in \mathbb{N}$ such that $|a_n| \leq C \frac{1}{n^{1+\epsilon}}$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is absolutely convergent;
2. if there exist $C > 0$ and $m \in \mathbb{N}$ such that $|a_n| \geq C \frac{1}{n}$ for all $n \geq m$, then $\sum_{n=0}^{\infty} a_n$ is either conditionally convergent or divergent.

Theorem 62.7 (ratio test for absolute convergence).

Let $\sum_{n=0}^{\infty} a_n$ with $a_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$.

1. If $L < 1$ then $\sum_{n=0}^{\infty} a_n$ is absolutely convergent and so is convergent.
2. If $L > 1$ or $L = \infty$ then $\sum_{n=0}^{\infty} a_n$ is divergent.
3. If $L = 1$ then the test fails.

Theorem 62.8 (root test for absolute convergence).

Let $\sum_{n=0}^{\infty} a_n$ with $a_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = L$.

1. If $L < 1$ then $\sum_{n=0}^{\infty} a_n$ is absolutely convergent and so is convergent.
2. If $L > 1$ or $L = \infty$ then $\sum_{n=0}^{\infty} a_n$ is divergent.
3. If $L = 1$ then the test fails.

63 Alternating series, Dirichlet's test and Abel's test

Definition 63.1.

A series of the form $\sum_{n=0}^{\infty} (-1)^n a_n$ or $\sum_{n=0}^{\infty} (-1)^{n+1} a_n$ with $a_n > 0$ is called an **alternating series**.

Theorem 63.2 (alternating series test).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers.

If $0 < a_{n+1} \leq a_n$ for all $n \geq 0$ and $\lim_{n \rightarrow \infty} a_n = 0$, then

1. the alternating series $\sum_{n=0}^{\infty} (-1)^n a_n$ is convergent;
2. $\sum_{n=0}^m (-1)^n a_n \geq \sum_{n=0}^{\infty} (-1)^n a_n$ for all even $m \geq 0$,
 $\sum_{n=0}^m (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n$ for all odd $m > 0$, and
 $|\sum_{n=0}^{\infty} (-1)^n a_n - \sum_{n=0}^m (-1)^n a_n| \leq a_{m+1}$ for all $m \geq 0$.

Theorem 63.3 (Dirichlet's test for convergence of series).

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers and $\{b_n\}_{n=0}^{\infty}$ be a sequence of complex numbers such that $0 < a_{n+1} \leq a_n$ for all $n \geq 0$, $\lim_{n \rightarrow \infty} a_n = 0$,

and there exists $C > 0$ such that $|\sum_{n=0}^m b_n| \leq C$ for all $m \geq 0$.

Then the series $\sum_{n=0}^{\infty} a_n b_n$ is convergent.

Corollary 63.4.

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers

with $0 < a_{n+1} \leq a_n$ for all $n \geq 0$ and $\lim_{n \rightarrow \infty} a_n = 0$.

Then $\sum_{n=0}^{\infty} a_n \sin n\theta$ is convergent for all $\theta \in \mathbb{R}$

and $\sum_{n=0}^{\infty} a_n \cos n\theta$ is convergent for all $\theta \in \mathbb{R} \setminus \{2k\pi : k \in \mathbb{Z}\}$.

Theorem 63.5 (Abel's test for convergence of series).

Let $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$ be two sequences of real numbers.

Suppose that $0 < a_{n+1} \leq a_n$ for all $n \geq 0$ and $\sum_{n=0}^{\infty} b_n$ is convergent.

Then the series $\sum_{n=0}^{\infty} a_n b_n$ is convergent.

64 Double series and the product of two series

Definition 64.1.

A **double series** is an expression of the form $\sum_{m,n=0}^{\infty} a_{mn}$, where $a_{mn} \in \mathbb{R}$ for each ordered pair $mn := (m, n)$ with $m, n \in \mathbb{N} \cup \{0\}$, that is, a series whose terms are indexed by ordered pairs of nonnegative integers. If for each integer $m \geq 0$, the series $\sum_{n=0}^{\infty} a_{mn}$ is convergent, then we call the series $\sum_{m=0}^{\infty} (\sum_{n=0}^{\infty} a_{mn})$ the **iterated series**, no matter it is convergent or not.

For a double series $\sum_{m,n=0}^{\infty} a_{mn}$, let $\sigma : \mathbb{N} \cup \{0\} \rightarrow \{mn := (m, n) : m, n \in \mathbb{N} \cup \{0\}\}$ be a one-to-one and onto map and let $b_i = a_{\sigma(i)}$.

We call such an ordinary series $\sum_{i=0}^{\infty} b_i$ an **ordering** of the double series $\sum_{m,n=0}^{\infty} a_{mn}$.

We say that the double series $\sum_{m,n=0}^{\infty} a_{mn}$ is **absolutely convergent** if an ordering $\sum_{i=0}^{\infty} b_i$ of $\sum_{m,n=0}^{\infty} a_{mn}$ is absolutely convergent; in this case, we write $\sum_{m,n=0}^{\infty} a_{mn} = \sum_{i=0}^{\infty} b_i$ and call it the **sum** of $\sum_{m,n=0}^{\infty} a_{mn}$.

Proposition 64.2.

Let $\sum_{m,n=0}^{\infty} a_{mn}$ be a double series. We have the following:

1. If an ordering of $\sum_{m,n=0}^{\infty} a_{mn}$ is absolutely convergent, then all orderings of $\sum_{m,n=0}^{\infty} a_{mn}$ are absolutely convergent and have the same sum.
2. The double series $\sum_{m,n=0}^{\infty} a_{mn}$ is absolutely convergent $\iff$ the iterated series $\sum_{m=0}^{\infty} (\sum_{n=0}^{\infty} |a_{mn}|)$ is convergent, in which case, $\sum_{m,n=0}^{\infty} a_{mn} = \sum_{m=0}^{\infty} (\sum_{n=0}^{\infty} a_{mn})$.
3. If the double series $\sum_{m,n=0}^{\infty} a_{mn}$ is not absolutely convergent, then we have the following cases:
 - (a) If $\sum_{m,n=0}^{\infty} \max\{a_{mn}, 0\} = +\infty$ and $\sum_{m,n=0}^{\infty} \max\{-a_{mn}, 0\} < +\infty$, then all orderings of $\sum_{m,n=0}^{\infty} a_{mn}$ are divergent with sum $+\infty$.
 - (b) If $\sum_{m,n=0}^{\infty} \max\{a_{mn}, 0\} < +\infty$ and $\sum_{m,n=0}^{\infty} \max\{-a_{mn}, 0\} = +\infty$, then all orderings of $\sum_{m,n=0}^{\infty} a_{mn}$ are divergent with sum $-\infty$.
 - (c) If $\sum_{m,n=0}^{\infty} \max\{a_{mn}, 0\} = \sum_{m,n=0}^{\infty} \max\{-a_{mn}, 0\} = +\infty$ and $\lim_{m,n \rightarrow \infty} a_{mn} = 0$, then various ordering of $\sum_{m,n=0}^{\infty} a_{mn}$ can converge to any real number.

Theorem 64.3 (absolute convergence of the product of two series).

If $\sum_{m=0}^{\infty} a_m$ and $\sum_{n=0}^{\infty} b_n$ are both absolutely convergent with sums A and B respectively, then $\sum_{m,n=0}^{\infty} a_m b_n$ is absolutely convergent and its sum is AB .

Theorem 64.4.

Let $\sum_{m=0}^{\infty} a_m$ and $\sum_{n=0}^{\infty} b_n$ be both convergent with sums A and B respectively. Then

1. if at least one of $\sum_{m=0}^{\infty} a_m$ and $\sum_{n=0}^{\infty} b_n$ is absolutely convergent, then $\sum_{k=0}^{\infty} \left(\sum_{\substack{m+n=k \\ m,n \geq 0}} a_m b_n \right)$, called the **Cauchy product** of $\sum_{m=0}^{\infty} a_m$ and $\sum_{n=0}^{\infty} b_n$, is convergent with sum AB ;
2. if both of $\sum_{m=0}^{\infty} a_m$ and $\sum_{n=0}^{\infty} b_n$ are conditionally convergent, then their Cauchy product may be divergent.

65 Sequences of functions

Definition 65.1.

Let $S \subseteq \mathbb{R}^k$, let $f : S \rightarrow \mathbb{R}$ be a function,
and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

1. We say that the sequence $\{f_n\}_{n=0}^\infty$ of functions is **pointwise convergent on S to f** ,
if for any $x \in S$, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$,
that is, for any $x \in S$ and for any $\epsilon > 0$, there exists $N \in \mathbb{N}$
such that if $n \geq N$ then $|f_n(x) - f(x)| < \epsilon$;
in which case, we write $f_n \rightarrow f$ **pointwise on S** (as $n \rightarrow \infty$),
and we use $\lim_{n \rightarrow \infty} f_n = f$ as a notion of a function, called the **limit function** of $\{f_n\}_{n=0}^\infty$.
2. We say that the sequence $\{f_n\}_{n=0}^\infty$ of functions is **uniformly convergent on S to f** ,
if for any $\epsilon > 0$, there exists $N \in \mathbb{N}$
such that if $n \geq N$ and $x \in S$, then $|f_n(x) - f(x)| < \epsilon$;
in which case, we write $f_n \rightarrow f$ **uniformly on S** .
3. We say that the sequence $\{f_n\}_{n=0}^\infty$ of functions is **uniformly Cauchy on S** ,
if for any $\epsilon > 0$, there exists $N \in \mathbb{N}$
such that if $m, n \geq N$ and $x \in S$, then $|f_m(x) - f_n(x)| < \epsilon$;

Theorem 65.2.

Let $S \subseteq \mathbb{R}^k$, let $f : S \rightarrow \mathbb{R}$ be a function,
and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

Then $f_n \rightarrow f$ uniformly on S

$\iff$ there exist $N \in \mathbb{N}$ and a sequence $\{C_n\}_{n=N}^\infty$ of real numbers
such that $C_n \geq 0$ for all $n \geq N$, $\lim_{n \rightarrow \infty} C_n = 0$,
and $|f_n(x) - f(x)| \leq C_n$ for all $x \in S$ and all $n \geq N$.

Theorem 65.3.

Let $S \subseteq \mathbb{R}^k$ and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

Then $\{f_n\}_{n=0}^\infty$ is uniformly Cauchy on S

$\iff$ there exists a function $f : S \rightarrow \mathbb{R}$ such that $f_n \rightarrow f$ uniformly on S .

Theorem 65.4.

Let $S \subseteq \mathbb{R}^k$, let $f : S \rightarrow \mathbb{R}$ be a function,

and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

1. If $f_n \rightarrow f$ uniformly on S and the function f_n is continuous on S for each $n \geq 0$,
then f is continuous on S ,
that is, $\lim_{k \rightarrow \infty} f(x_k) = \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} f_n(x_k) = \lim_{n \rightarrow \infty} \lim_{k \rightarrow \infty} f_n(x_k) = \lim_{n \rightarrow \infty} f_n(x) = f(x)$, for any $\{x_k\}_{k=0}^\infty \subset S$ with $\lim_{k \rightarrow \infty} x_k = x \in S$.
2. If $k = 1$, $S = (a, b)$, $f_n \rightarrow f$ uniformly on $[a + \eta, b - \eta]$ for each $\eta > 0$, (not necessarily, $f_n \rightarrow f$ uniformly on (a, b)),
and f_n is continuous on (a, b) for each $n \geq 0$,
then f is continuous on (a, b) .

Theorem 65.5 (the Weierstrass approximation theorem).

Let f be a real-valued continuous function on $[a, b]$,

then there is a sequence $\{p_n\}_{n=0}^\infty$ of polynomials such that $p_n \rightarrow f$ uniformly on $[a, b]$.

Theorem 65.6 (a generalization of Dini's theorem).

Let $\{f_n\}_{n=0}^\infty$ be a sequence of real-valued continuous functions on a compact subset S of $\mathbb{R}^m$ such that

(i) for each $x \in S$, the sequence $\{f_n(x)\}_{n=0}^\infty$ is bounded and monotone, and

(ii) the function $x \mapsto \lim_{n \rightarrow \infty} f_n(x)$ is well defined and continuous on S .

Then $f_n \rightarrow \lim_{n \rightarrow \infty} f_n$ uniformly on S .

Remark that the result is not always true without the monotonicity of item (i).

Example 65.7.

For each $n \in \mathbb{N}$, define $f_n : \mathbb{R} \rightarrow \mathbb{R}$ by $f_n(x) = \frac{\arctan(nx)}{n}$.

Then $f'_n(x) = \frac{1}{n^2 x^2 + 1}$ and $f''_n(x) = \frac{-2nx}{(n^2 x^2 + 1)^2}$.

Moreover,

$$\begin{aligned} \lim_{n \rightarrow \infty} f'_n(0) &= 1 \neq 0 = \left(\lim_{n \rightarrow \infty} f_n \right)'(0), \\ \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} f'_n(x) &= 1 \neq 0 = \lim_{x \rightarrow 0} \lim_{n \rightarrow \infty} f'_n(x), \\ \lim_{n \rightarrow \infty} \int_0^1 f''_n(x) dx &= 1 \neq 0 = \int_0^1 \lim_{n \rightarrow \infty} f''_n(x) dx \end{aligned}$$

66 Series of functions

Definition 66.1.

Let $S \subseteq \mathbb{R}^k$ and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

1. We say that the series $\sum_{n=0}^{\infty} f_n$ of functions is **pointwise convergent** on S , if for each $x \in S$, the series $\sum_{n=0}^{\infty} f_n(x)$ is convergent; in this case, the notation $\sum_{n=0}^{\infty} f_n$ also denotes the function from S to $\mathbb{R}$ defined by $(\sum_{n=0}^{\infty} f_n)(x) := \sum_{n=0}^{\infty} f_n(x)$.
2. We say that the series $\sum_{n=0}^{\infty} f_n$ of functions is **absolutely convergent** on S , if for each $x \in S$, the series $\sum_{n=0}^{\infty} f_n(x)$ is absolutely convergent.
3. We say that the series $\sum_{n=0}^{\infty} f_n$ of functions is **uniformly convergent** on S , if the series $\sum_{n=0}^{\infty} f_n$ of functions is pointwise convergent on S , and the sequence $\{\sum_{n=0}^m f_n : S \rightarrow \mathbb{R}\}_{m=0}^{\infty}$ of functions, defined by $(\sum_{n=0}^m f_n)(x) = \sum_{n=0}^m f_n(x)$, is uniformly convergent to the function $\sum_{n=0}^{\infty} f_n$ on S .

Theorem 66.2 (the Weierstrass test/the Weierstrass M test).

Let $S \subseteq \mathbb{R}^k$ and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

If there exists a sequence $\{M_n\}_{n=0}^{\infty}$ of real numbers

such that $M_n \geq 0$ for all $n \geq 0$, $\sum_{n=0}^{\infty} M_n$ is convergent, and $|f_n(x)| \leq M_n$ for all $x \in S$ and all $n \geq 0$, then the series $\sum_{n=0}^{\infty} f_n$ is absolutely convergent on S and is uniformly convergent on S .

Theorem 66.3.

Let $S \subseteq \mathbb{R}^k$ and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

1. If the series $\sum_{n=0}^{\infty} f_n$ is uniformly convergent on S and f_n is continuous on S for each $n \geq 0$, then the function $\sum_{n=0}^{\infty} f_n : S \rightarrow \mathbb{R}$ is continuous on S .
2. If $k = 1$, $S = (a, b)$, the series $\sum_{n=0}^{\infty} f_n$ is uniformly convergent on $[a + \eta, b - \eta]$ for any $\eta > 0$, and f_n is continuous on (a, b) for each $n \geq 0$, then the function $\sum_{n=0}^{\infty} f_n : (a, b) \rightarrow \mathbb{R}$ is continuous on (a, b) .

Theorem 66.4 (Cauchy condition for uniform convergence of series).

Let $S \subseteq \mathbb{R}^k$ and for each $n \in \mathbb{N} \cup \{0\}$, let $f_n : S \rightarrow \mathbb{R}$ be a function.

Then the series $\sum_{n=0}^{\infty} f_n$ is uniformly convergent on S .

$\iff$ for any $\varepsilon > 0$, there exist $N \in \mathbb{N}$ such that if $m \geq N$, $p \geq 0$, and $x \in S$, then $|\sum_{n=m}^{m+p} f_n(x)| < \varepsilon$.

67 Integrals and derivatives of sequences and series

Theorem 67.1 (interchange of limit and integral).

Let $S \subseteq \mathbb{R}^k$ be Jordan measurable, $f : S \rightarrow \mathbb{R}$ be Riemann integrable on S , and for each $n \in \mathbb{N} \cup \{0\}$, $f_n : S \rightarrow \mathbb{R}$ be Riemann integrable on S .

If $f_n \rightarrow f$ uniformly on S , then $\lim_{n \rightarrow \infty} \int_S f_n(x) d^k x = \int_S f(x) d^k x$.

Theorem 67.2 (interchange of limit and derivative).

1. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function, $g : [a, b] \rightarrow \mathbb{R}$ be a function, and for each $n \in \mathbb{N} \cup \{0\}$, $f_n : [a, b] \rightarrow \mathbb{R}$ be C^1 on $[a, b]$.
If $f_n \rightarrow f$ pointwise on $[a, b]$ and $f'_n \rightarrow g$ uniformly on $[a, b]$, then f is C^1 on $[a, b]$ and $f'(x) = g(x)$ for all $x \in [a, b]$.
2. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function and for each $n \in \mathbb{N} \cup \{0\}$, $f_n : [a, b] \rightarrow \mathbb{R}$ be C^1 on $[a, b]$.
If $f_n \rightarrow f$ pointwise on $[a, b]$ and $\{f'_n\}_{n=0}^\infty$ is uniformly Cauchy on $[a, b]$, then f is C^1 on $[a, b]$ and $f'_n \rightarrow f'$ uniformly on $[a, b]$; and hence $\lim_{n \rightarrow \infty} f'_n = (\lim_{n \rightarrow \infty} f_n)'$ whenever we write $\lim_{n \rightarrow \infty} f_n = f$.
3. Let $f : (a, b) \rightarrow \mathbb{R}$ be a function, $g : (a, b) \rightarrow \mathbb{R}$ be a function, and for each $n \in \mathbb{N} \cup \{0\}$, $f_n : (a, b) \rightarrow \mathbb{R}$ be C^1 on (a, b) .
If $f_n \rightarrow f$ pointwise on (a, b) and $f'_n \rightarrow g$ uniformly on $[a + \eta, b - \eta]$ for any $\eta > 0$, then f is C^1 on (a, b) and $f'(x) = g(x)$ for all $x \in (a, b)$.

Theorem 67.3 (interchanges of series and integral, and series and derivative).

For each $n \in \mathbb{N} \cup \{0\}$, let $f_n : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$

such that the series $\sum_{n=0}^\infty f_n$ is pointwise convergent on $[a, b]$.

1. If the series $\sum_{n=0}^\infty f_n$ is uniformly convergent on $[a, b]$,
then $\int_a^b \sum_{n=0}^\infty f_n(x) dx = \sum_{n=0}^\infty \int_a^b f_n(x) dx$.
2. If for each $n \in \mathbb{N} \cup \{0\}$, $f_n : [a, b] \rightarrow \mathbb{R}$ is C^1 on $[a, b]$,
and the series $\sum_{n=0}^\infty f'_n$ is uniformly convergent on $[a, b]$,
then the function $\sum_{n=0}^\infty f_n$ is C^1 on $[a, b]$
and $\frac{d}{dx} \sum_{n=0}^\infty f_n(x) = \sum_{n=0}^\infty \frac{d}{dx} f_n(x)$ for all $x \in [a, b]$.

68 Power series

Definition 68.1.

A series of the form $\sum_{n=0}^{\infty} a_n x^n$, where x is a real or complex variable and a_n 's are constants, is called a **power series** in x .

The constants a_n 's are called the **coefficients** of the power series $\sum_{n=0}^{\infty} a_n x^n$.

Assigning a fixed x , the power series $\sum_{n=0}^{\infty} a_n x^n$ is a series of the sequence $\{a_n x^n\}_{n=0}^{\infty}$, which might be convergent or divergent.

Given a constant c , a series of the form $\sum_{n=0}^{\infty} a_n (x - c)^n$ is called a **power series** in $x - c$.

Theorem 68.2.

Let $x_0, x_1 \in \mathbb{R}$ with $x_0 \neq 0$ and $x_1 \neq 0$.

1. If a power series $\sum_{n=0}^{\infty} a_n x^n$ is convergent for $x = x_0$, then the series $\sum_{n=0}^{\infty} a_n x^n$ is absolutely convergent for all $x \in \mathbb{R}$ with $|x| < |x_0|$.
2. If a power series $\sum_{n=0}^{\infty} a_n x^n$ is divergent for $x = x_1$, then the series $\sum_{n=0}^{\infty} a_n x^n$ is divergent for all $x \in \mathbb{R}$ with $|x| > |x_1|$.
3. For any power series $\sum_{n=0}^{\infty} a_n (x - c)^n$, there exists a number $r \in [0, \infty]$, called the **radius of convergence** for the power series $\sum_{n=0}^{\infty} a_n (x - c)^n$, such that the series $\sum_{n=0}^{\infty} a_n (x - c)^n$ is absolutely convergent for all $x \in \{z : |z - c| < r\}$ and is divergent for all $x \in \{z : |z - c| > r\}$.
Moreover, the radius of convergence for the power series $\sum_{n=0}^{\infty} a_n (x - c)^n$ is the reciprocal of $\limsup_{n \rightarrow \infty} |a_n|^{1/n}$ which can be replaced by $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ if the latter exists.

Theorem 68.3.

If r is the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$, then for any $0 \leq r_0 < r$, the series $\sum_{n=0}^{\infty} a_n x^n$ of functions $x \mapsto a_n x^n$ is uniformly convergent on the set $\{x : |x| \leq r_0\}$ and the function $x \mapsto \sum_{n=0}^{\infty} a_n x^n$ is a continuous on $\{x : |x| < r\}$.

Theorem 68.4.

Let $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ be power series with radii of convergence r_a and r_b , resp. Then

1. if $c \in \mathbb{R}$ then $c \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} c a_n x^n$ for all $|x| < r_a$;
2. $\sum_{n=0}^{\infty} a_n x^n \pm \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n \pm b_n) x^n$ for all $|x| < \min\{r_a, r_b\}$;
3. $\left(\sum_{n=0}^{\infty} a_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n$ for all $|x| < \min\{r_a, r_b\}$.

Proposition 68.5.

Consider a rational function $f(x) = \frac{q(x)}{p(x)}$, where $p(x)$ and $q(x)$ are polynomials.

If $p(x_0) \neq 0$ for some $x_0 \in \mathbb{R}$, then $f(x)$ is C^∞ on an open interval containing x_0 .

If, in addition, all common factors of $p(x)$ and $q(x)$ have been cancelled,

then $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ for $x \in (x_0 - r, x_0 + r)$,

and r is the radius of convergence for the power series,

where $r = \min\{|z - x_0| : z \in \mathbb{C} \text{ such that } p(z) = 0\}$.

The proof can be done by using Cauchy's integral formula in complex analysis;

refer to Boas's book, p61.

69 Differentiation and integration of power series, and Abel's theorem, and the Arzela-Ascoli theorem

Theorem 69.1 (differentiation and integration of power series).

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $r > 0$

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for all $x \in (-r, r)$.

(In this case, we call $\sum_{n=0}^{\infty} a_n x^n$ a **power series representation of f on $(-r, r)$** .)

Then we have the followings:

1. The number r is the radius of convergence for the power series $\sum_{n=1}^{\infty} n a_n x^{n-1}$, which is called the **derivative series** of $\sum_{n=0}^{\infty} a_n x^n$.
2. The function f is differentiable at $x \in (-r, r)$ and $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ for all $x \in (-r, r)$.
3. The function f is C^∞ on the interval $(-r, r)$ and its k -th derivative may be computed on $(-r, r)$ by differentiating the series $\sum_{n=0}^{\infty} a_n x^n$ termwise k times.
4. If $-r < a < b < r$, then $\int_a^b f(t) dt = \sum_{n=0}^{\infty} a_n \frac{b^{n+1} - a^{n+1}}{n+1}$.
In particular, $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ for $0 \leq x < r$;
in this case, we call $\sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ the **integral series** of $\sum_{n=0}^{\infty} a_n x^n$.

Corollary 69.2.

Let $r > 0$ be the radius of convergence for the power series $\sum_{n=0}^{\infty} a_n x^n$.

1. If $f : (-r, r) \rightarrow \mathbb{R}$ is a function such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < r$, then $a_n = \frac{f^{(n)}(0)}{n!}$ for all $n \geq 0$.
That is, every power series with a positive radius of convergence is the Taylor series of its sum.
2. If $\sum_{n=0}^{\infty} b_n x^n$ is a power series such that $\sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} a_n x^n$ for $|x| < r$, then $b_n = a_n$ for all $n \geq 0$.

Theorem 69.3 (Abel's theorem).

If the series $\sum_{n=0}^{\infty} a_n x^n$ is convergent at $x = r$ (resp. $x = -r$),

then the series $\sum_{n=0}^{\infty} a_n x^n$ of functions $x \mapsto a_n x^n$ is uniformly convergent

on the interval $[0, r]$ (resp. $[-r, 0]$)

and hence the function $x \mapsto \sum_{n=0}^{\infty} a_n x^n$ is continuous on the interval $[0, r]$ (resp. $[-r, 0]$).

Corollary 69.4.

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers.

If $\sum_{n=0}^{\infty} a_n$ is convergent, then $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n$ exists and is equal to $\sum_{n=0}^{\infty} a_n$.

However, it may occur that $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n$ exists but $\sum_{n=0}^{\infty} a_n$ is divergent.

In this case, the divergent series $\sum_{n=0}^{\infty} a_n$ is said to

be **Abel summable** to the sum $\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n$.

Theorem 69.5 (the Arzela-Ascoli theorem, the Heine-Borel theorem for functional spaces).

Let S be a compact subset of $\mathbb{R}^k$ and $C^0(S, \mathbb{R}^m)$ be the set of all continuous functions from S to $\mathbb{R}^m$

equipped with the metric $d(f, g) = \sup\{\|f(x) - g(x)\| : x \in S\}$, called the C^0 -sup-norm metric.

Then a subset $\mathfrak{F}$ of $C^0(S, \mathbb{R}^m)$ is compact if and only if $\mathfrak{F}$ is closed, bounded, and (**uniformly**) **equicontinuous**

(that is, for any $\epsilon > 0$ there exists $\delta > 0$ such that

if $x, y \in S$ with $\|x - y\| < \delta$ and $f \in \mathfrak{F}$, then $\|f(x) - f(y)\| < \epsilon$,

which is, due to the compactness of S , equivalent to (**pointwise**) **equicontinuous**

that is, for any $x_0 \in S$ and any $\epsilon > 0$ there exists $\delta > 0$ such that

if $x \in S$ with $\|x - x_0\| < \delta$ and $f \in \mathfrak{F}$, then $\|f(x) - f(x_0)\| < \epsilon$).